

A scenic landscape featuring a stone path that leads from the foreground into a lush green field. In the background, there are rolling hills under a sky with scattered clouds. The path is made of flat, grey stones and is flanked by tall grass and some small white flowers. The overall atmosphere is peaceful and open.

IT Career Paths and Project Use Cases

Exploring Different IT Fields and Advancing Your Career with Purpose

Agenda



1. Introduction
2. My Journey in IT
3. Building your skillset
4. Preparing for your IT Career
5. Certificates
6. Additional Recommendations
7. Use Cases

Introduction



Careers in IT are versatile. Choose a path that makes you happy and comfortable, as it will be your daily job. Regularly upskill to maintain a strong professional status.



By the end of this Session you will have an overview of a career path in IT and how easily you can shift your goals

My Journey in IT



University Degree



Career options



Researching the market



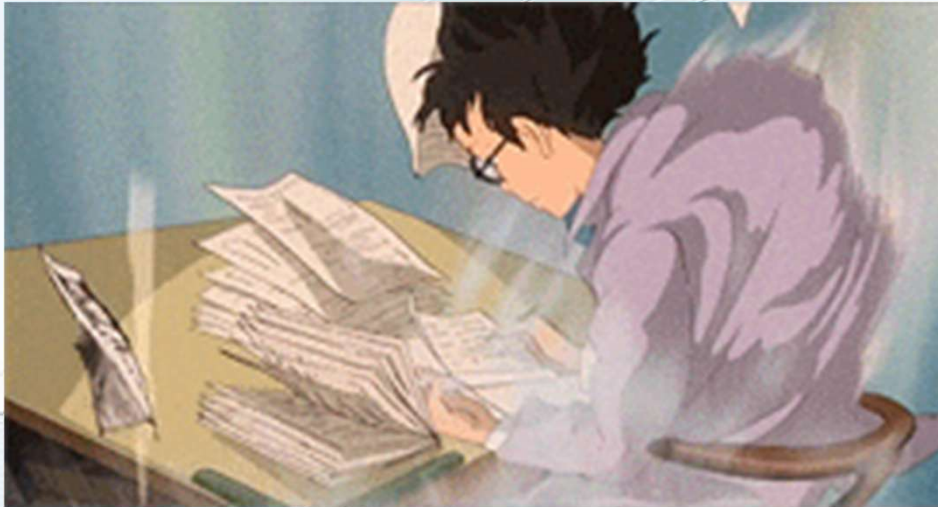
Technical support



Operation and on-site support



Consultancy –
Solution Design
and delivery



Building Your Skill Set

Build your knowledge base:

- **Computer Networks**
- **Scripting: Basics of Python**
- **Network Tools: Wireshark**
- **Operating Systems and services**
- **EVE-NG network Emulators**



Preparing for Your IT Career



Your Efforts will always
lead you to the right path



Obtain Your Certificates



Self-Study Certifications
Are a must

Obtain Your Certificates



Cisco (CCNA)



Microsoft (AZURE
Fundamentals)



CCNP (fields R&S, SP,
SDWAN, EI, DC)



CCIE (EI, SP, Security,
Wireless,
Collaboration, Devnet)



Project management,
Program Management

Additional Recommendations



Understand the different fields of IT careers



Lookout for new cutting-edge technology news

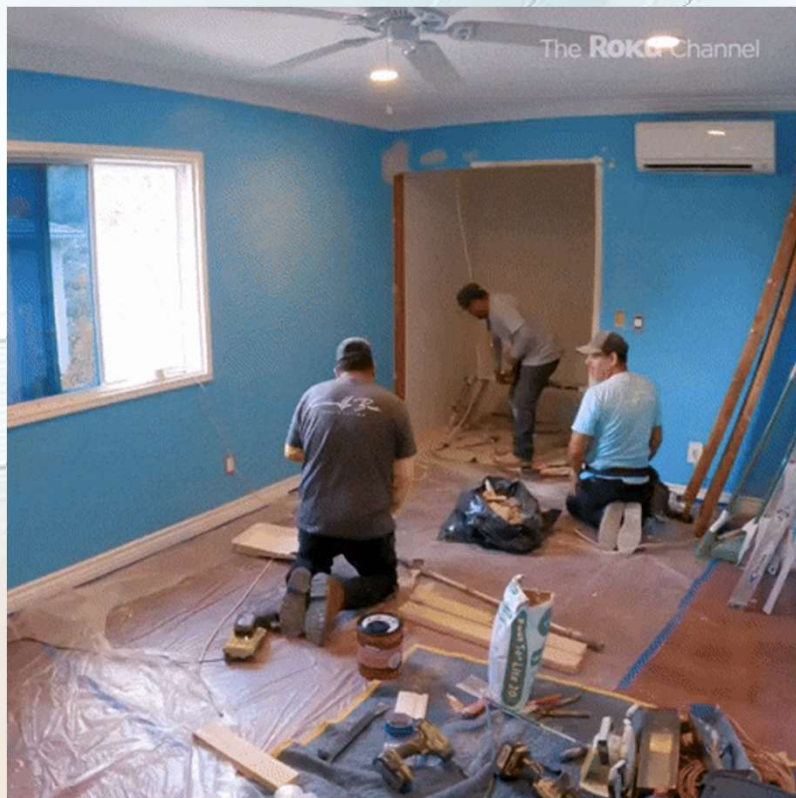


Check with your fellow students and older students about the opportunities they aim for



Conversations lead to knowledge sharing and creates allies

Behind the Scenes of Telecom Migrations



How Networks Evolve Without
Breaking the Internet

What Is a Migration in SP Networks?



- Moving live services from one platform to another
- Ensuring continuity while upgrading technology
- Balancing risk, timing, and customer impact

Why Do SPs Migrate?



- Aging hardware
- Vendor transitions (Huawei → Cisco)
- New technologies
- Cost, performance, security

Story Time: Huawei to Cisco Migration



Old network: Huawei
Device



New network: Cisco
infrastructure



Goal: Modernize
without downtime
Security Compliance

Migration Workflow

Discovery

Gap
Analysis

Design

MOP
Creation

Scheduling the Maintenance Window



- Low-traffic hours
- Communication plan
- Pre-checks

MW Night: The Plot Twist



- Feature mismatches
- Legacy configs
- Hidden dependencies
- Engineers adapt in real time

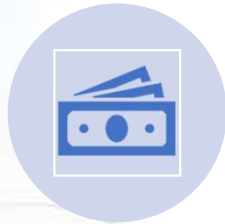
Story 2 Cellular Migration: RTN → NCS540



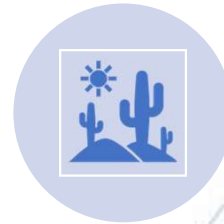
Why Operators Used RTN for Years



FAST TO DEPLOY IN
REMOTE AREAS



CHEAPER THAN LAYING
FIBER



WORKS WELL ACROSS
MOUNTAINS, DESERTS,
RURAL ZONES



RELIABLE WHEN
MAINTAINED PROPERLY

Quick 2g 3g 4g 5g intro



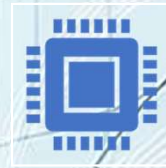
2G — The Voice Era



**3G — The Mobile
Internet Begins**



**4G — The
Broadband Era**



**5G — The Ultra-Fast,
Low-Latency Era**

What Is a Government MW Wave?

- Licensed microwave frequency
- High reliability
- Strict SLAs
- operators often lease spectrum from government entities.



Migration Steps



Identify RTN links



Validate MW wave capacity



Build parallel NCS540 path



Move traffic generation by generation



Challenges in Cellular Migration



Timing stability



Microwave
bandwidth limits



Fiber Reach



Shelter location



Redundancy

MPLS Story

MPLS = postal system
with labels

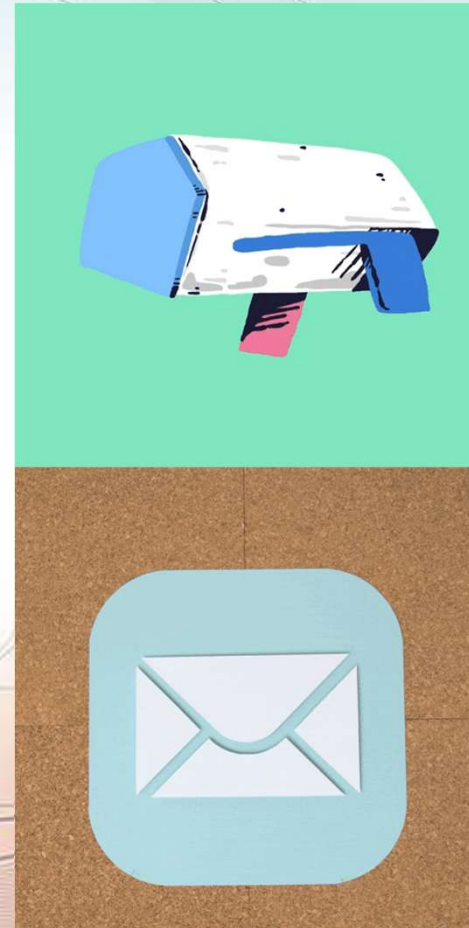
LDP = old sorting
method

SR = smart routing
with built-in directions



LDP → SR Migration (Zero Downtime)

- Run LDP + SR in parallel
- Gradually shift services
- Remove LDP after full migration





Q&A

